# Urinary anion gap — Q61

## Calculate the urinary anion gap and explain what it measures

- UAG = urine (Na + K) − Cl = (40 + 30) − 50 = +20 mmol/L
- UAG is an indirect measure of urinary ammonium (NH4+) excretion (NH4+ is excreted with Cl−)
- Negative UAG (−20 to −50) = high NH4+ excretion → kidneys responding normally → GI bicarbonate loss (diarrhoea)
- Positive UAG = low NH4+ → renal cause (distal RTA)

## Interpret her results and give the most likely diagnosis

- Positive UAG + urine pH > 5.5 despite acidosis + hypokalaemia
- Distal (type 1) renal tubular acidosis
- Causes: Sjögren, SLE, amphotericin, lithium, toluene; complications: nephrocalcinosis, stones

## What are the limitations of the urinary anion gap, and what is the urine osmolar gap

- Unreliable if other unmeasured anions in urine (ketones, hippurate from toluene, drugs like penicillins)
- Unreliable if urine Na < 20 (volume depletion) or polyuria
- Urine osmolar gap = measured urine osm − [2(Na + K) + urea + glucose]; NH4+ ≈ gap/2
- Urine osmolar gap > 100–150 mOsm/kg = adequate NH4+ excretion

# Urinary anion gap — Q62

## Compare the three main types of renal tubular acidosis

- Type 1 (distal): impaired H+ secretion; urine pH > 5.5; low K; stones/nephrocalcinosis; positive UAG
- Type 2 (proximal): impaired HCO3 reabsorption; urine pH < 5.5 when HCO3 low; low K; Fanconi syndrome (glycosuria, phosphaturia), myeloma, acetazolamide, tenofovir
- Type 4: hypoaldosteronism or resistance; high K; urine pH < 5.5; mild acidosis
- Type 4 causes: diabetic nephropathy (hyporeninaemic hypoaldosteronism), ACEI/ARB, spironolactone, heparin, trimethoprim

## What is the diagnosis in this patient and how would you treat it

- Type 4 RTA (hyperkalaemia, urine pH < 5.5, diabetes, ACEI)
- Reduce/stop K-sparing drugs; low K diet; loop or thiazide diuretic
- Fludrocortisone if not hypertensive/fluid overloaded; potassium binder

## List the causes of a normal anion gap metabolic acidosis

- GI HCO3 loss: diarrhoea, ileostomy, pancreatic fistula, ureterosigmoidostomy
- Renal: RTA, carbonic anhydrase inhibitors
- Iatrogenic: large-volume 0.9% saline
- Early CKD, recovery from DKA (ketones lost in urine)

# Stewart approach and SID — Q63

## Describe the three independent variables of the Stewart approach

- Strong ion difference (SID): fully dissociated cations minus anions (Na + K + Ca + Mg − Cl − lactate); normal ~40–42
- Total weak acids (Atot): mainly albumin and phosphate
- PaCO2
- H+ and HCO3− are dependent variables

## Calculate the apparent SID and explain why large-volume saline causes acidosis

- SIDa ≈ Na + K − Cl − lactate = 142 + 4 − 120 − 1 = 25 mEq/L (low; normal ~40)
- Low SID → more water dissociation → higher H+ → acidosis
- 0.9% saline has Na = Cl = 154 → SID 0; large volumes dilute plasma SID toward zero
- Balanced solutions have an effective SID ~27–29 → less acidosis

## How does his low albumin affect the acid–base picture

- Albumin is a weak acid; hypoalbuminaemia is a metabolic alkalosis
- It masks the acidosis and lowers the anion gap (correct AG: + 2.5 for every 10 g/L fall in albumin)
- Strong ion gap (SIG = SIDa − SIDe) identifies unmeasured anions

# Stewart approach and SID — Q64

**Using the traditional approach, calculate the albumin-corrected anion gap and the delta ratio**

- AG = 138 − (90 + 24) = 24 mmol/L
- Corrected AG = 24 + 2.5 × (40 − 20)/10 = 24 + 5 = 29 mmol/L
- Delta: AG has risen by ~17 but HCO3 has not fallen at all (ΔAG/ΔHCO3 ≫ 2) → coexisting metabolic alkalosis
- Mixed disorder: high AG (lactic) acidosis + metabolic alkalosis (vomiting) with normal pH

**Interpret the same results using the Stewart approach**

- SIDa = 138 − 90 − 6 = 42 (normal) despite lactate – because low chloride (vomiting) raises SID
- Hypoalbuminaemia (low Atot) adds alkalosis
- Lactate acidosis is hidden by hypochloraemic and hypoalbuminaemic alkalosis

**Compare the advantages and limitations of the bicarbonate, base excess and Stewart approaches**

- Bicarbonate/AG: simple, bedside; misses mixed disorders if albumin not corrected
- Base excess: quantifies metabolic component; does not explain cause; partitioned BE helps
- Stewart: explains mechanisms (chloride, albumin, unmeasured anions); more complex, needs many measurements

# Dialysis disequilibrium syndrome — Q65

## What is the diagnosis and its mechanism

- Dialysis disequilibrium syndrome (DDS)
- Rapid removal of urea from blood; urea leaves the brain more slowly (urea transporter down-regulation in uraemia)
- Osmotic gradient → water moves into brain cells → cerebral oedema
- Idiogenic osmoles and rapid correction of acidosis (CSF pH falls paradoxically) contribute

## List the risk factors

- First dialysis session with very high urea (> 60 mmol/L / BUN > 175 mg/dL)
- Severe metabolic acidosis; children and elderly
- Pre-existing neurological disease (TBI, stroke, seizures), raised ICP
- Aggressive (high blood flow, large dialyser, long) first session; hypernatraemia/hypoglycaemia

## How can it be prevented and treated

- First session short (2 h), low blood flow (150–250 mL/min), small dialyser – aim urea reduction ≤ 30–40%
- Daily short sessions or CRRT in high-risk patients (especially with brain injury)
- Treat: stop or slow dialysis; mannitol or hypertonic saline; seizure control
- Higher dialysate sodium

# Dialysis disequilibrium syndrome — Q66

## List the absolute indications for renal replacement therapy

- Refractory hyperkalaemia (> 6.5 mmol/L or ECG changes)
- Severe metabolic acidosis (pH < 7.15)
- Refractory fluid overload/pulmonary oedema
- Uraemic complications (pericarditis, encephalopathy, bleeding); dialysable toxins

## Summarise the evidence on timing of RRT in AKI

- AKIKI (2016) and STARRT-AKI (2020): early RRT does not reduce mortality vs a delayed strategy; more adverse events with early
- AKIKI 2 (2021): waiting too long (beyond the delayed trigger) is harmful
- Strategy: start RRT for an absolute indication or prolonged severe oliguria/high urea

## Compare CRRT and intermittent HD in this patient and choose a modality

- CRRT: slow continuous removal, better haemodynamic tolerance, steady fluid balance, less ICP rise
- IHD: rapid clearance (better for hyperkalaemia, toxins), allows mobilisation, cheaper
- No mortality difference in trials; CRRT/SLED preferred in shock or brain injury
- Choose CRRT (on vasopressors)
- Dose 20–25 mL/kg/h effluent (RENAL, ATN trials: higher dose no benefit)

# Citrate toxicity — Q67

## Explain how regional citrate anticoagulation works

- Citrate infused pre-filter chelates ionised calcium → prevents clotting in the circuit (target post-filter iCa 0.25–0.35 mmol/L)
- Most calcium–citrate complex is removed by the filter
- Remaining citrate is metabolised by liver (Krebs cycle) to bicarbonate; calcium is replaced post-filter to keep systemic iCa 1.0–1.2

## Calculate the total-to-ionised calcium ratio and interpret it

- Ratio = 3.1 / 0.85 ≈ 3.6
- Ratio > 2.5 indicates citrate accumulation
- Due to impaired citrate metabolism (liver failure, shock with poor perfusion) → bound calcium rises, ionised falls, metabolic acidosis with high AG

## How would you manage it

- Reduce or stop citrate (switch to heparin or no anticoagulation)
- Increase dialysate/replacement flow to remove more citrate–calcium complex
- Increase calcium replacement to correct ionised hypocalcaemia
- Treat acidosis; monitor every 4–6 h
- Note the opposite problem, citrate overload with good metabolism → metabolic alkalosis

# Citrate toxicity — Q68

## What is the most likely cause and its mechanism

- Citrate-induced hypocalcaemia
- Stored blood products contain citrate, which chelates ionised calcium
- During rapid transfusion (or with liver dysfunction/hypothermia) citrate metabolism is overwhelmed

## Describe the consequences of hypocalcaemia in massive transfusion

- Myocardial depression, hypotension, prolonged QT, arrhythmias
- Impaired coagulation (calcium is factor IV)
- Part of the 'lethal diamond': hypothermia, acidosis, coagulopathy, hypocalcaemia
- Neuromuscular irritability (paraesthesia, tetany)

## How would you monitor and treat it

- Monitor ionised calcium frequently (every 30–60 min) during massive transfusion
- Keep iCa > 1.1 mmol/L
- Calcium chloride 10% 10 mL (or gluconate 30 mL) IV, repeated as needed
- Warm products and patient; correct acidosis
- Viscoelastic-guided product replacement

# Severe hyponatraemia with seizure — Q69

## What is your immediate treatment? Give the dose and target

- 3% hypertonic saline 150 mL IV over 20 min (or 100 mL bolus; 2 mL/kg)
- Check Na after 20 min; repeat up to 2–3 times until symptoms stop
- Target rise 4–6 mmol/L in the first hours (enough to reduce cerebral oedema)
- Limit correction to ≤ 10 mmol/L in 24 h (8 if high risk of ODS) and ≤ 18 in 48 h

## Explain the pathophysiology of hyponatraemic encephalopathy

- Low plasma osmolality → water moves into brain cells → cerebral oedema
- Acute (< 48 h): no time for brain adaptation (loss of electrolytes and organic osmolytes) → high risk of herniation
- Exercise-associated: excess water intake + non-osmotic ADH release

## List the risk factors for osmotic demyelination syndrome

- Chronic hyponatraemia (> 48 h) with Na ≤ 105–110 mmol/L
- Alcoholism, malnutrition, liver disease
- Hypokalaemia
- Rapid correction (> 8–10 mmol/L/24 h)

Board 2026 recall · Severe hyponatraemia with seizure · Spasovski G et al. European clinical practice guideline on hyponatraemia. Eur J Endocrinol 2014

# Severe hyponatraemia with seizure — Q70

## What has happened and why is it dangerous

- Overcorrection (11 mmol/L in 12 h) due to water diuresis once volume restored (ADH switched off) and K replaced
- High risk of osmotic demyelination syndrome (chronic, alcoholic, hypokalaemic)
- ODS: dysarthria, dysphagia, quadriparesis, locked-in syndrome, days after correction

## How would you manage the overcorrection

- Stop hypertonic/isotonic saline
- Give desmopressin 1–2 µg IV/SC to stop the water diuresis (DDAVP clamp)
- Give 5% dextrose (e.g., 3 mL/kg/h) to lower Na back to the target
- Check Na every 1–2 h; aim to bring the rise back below 8 mmol/L/24 h

## Outline the diagnostic approach to hyponatraemia

- Measure plasma osmolality (exclude pseudohyponatraemia and hyperglycaemia)
- Urine osmolality: < 100 = excess water intake/beer potomania; > 100 = ADH present
- Urine Na: < 30 = low effective volume; > 30 = SIADH, diuretics, adrenal insufficiency, CSW
- Assess volume status; check TSH and cortisol

# Refeeding syndrome — Q71

### What is the most likely cause? Explain the pathophysiology

- Refeeding syndrome with severe hypophosphataemia
- Starvation depletes total body phosphate, K and Mg while serum levels may be normal
- Feeding (carbohydrate) → insulin release → intracellular shift of phosphate, K, Mg and glucose; phosphate used for ATP and 2,3-DPG
- Thiamine is consumed in carbohydrate metabolism → Wernicke's; sodium and water retention → fluid overload

### List the NICE criteria for high risk of refeeding

- One of: BMI < 16, unintentional weight loss > 15% in 3–6 months, little/no intake > 10 days, low K/PO4/Mg before feeding
- Or two of: BMI < 18.5, weight loss > 10%, little intake > 5 days, alcohol abuse or drugs (insulin, chemotherapy, antacids, diuretics)
- He is high risk (BMI 15, 10 days of no intake)

### Describe the clinical consequences of severe hypophosphataemia

- Respiratory muscle weakness, failure to wean
- Cardiac failure, arrhythmias
- Seizures, confusion, paraesthesia
- Rhabdomyolysis, haemolysis, white cell and platelet dysfunction

# Refeeding syndrome — Q72

## Outline your plan for starting nutrition safely

- Thiamine 200–300 mg daily (Pabrinex) before feeding and for 10 days, plus vitamin B complex and multivitamin
- Start low: 5–10 kcal/kg/day (5 if BMI < 14), increase slowly over 4–7 days to target
- Correct K, PO4, Mg before and during feeding
- Monitor electrolytes daily (or twice daily) for the first week; ECG monitoring

## Describe the replacement of phosphate

- Severe (< 0.3–0.5 mmol/L) or symptomatic: IV phosphate 0.3–0.5 mmol/kg over 6–12 h
- Use potassium or sodium phosphate depending on K
- Risks: hypocalcaemia, metastatic calcification, hyperkalaemia; caution in renal failure

## What does the evidence say about caloric restriction in ICU refeeding

- Doig 2015 (Refeeding RCT): in ICU refeeding hypophosphataemia, restricting calories to 20 kcal/h for 2 days improved survival
- Gradual return to full feeding over the next days
- Hypophosphataemia after starting feeding in ICU should trigger caloric restriction

Board 2026 recall · Refeeding syndrome · NICE CG32 2017